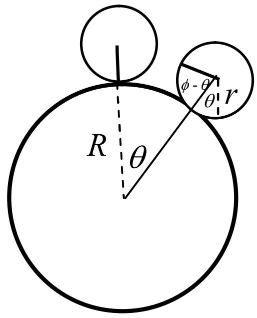
# *University Physics* Quiz 3-2

Name: ______________________ SID: ________________________ School/Department: __________________

1. A uniform sphere of mass $m$ and radius $r$ rolls without slipping on the outer surface of a fixed sphere of radius $R$. The position of the rolling sphere can be described by angle $\theta$ and its orientation can be described by angle $\phi$, as shown in the accompanying figure. The rolling sphere starts from the top of the fixed sphere ($\theta = 0$ and $\phi = 0$).




(a) What is the relation between $\theta$ and $\phi$?

(b) Find the velocity of the center of mass of the rolling sphere as a function of $\theta$ if the sphere starts from rest.

(c) Determine the value of $\theta$ at which the rolling sphere flies off the fixed sphere if it starts from rest.

(d) If the rolling sphere begins at time $t = 0$ with $\theta = 0$ but $\dot{\theta}(0) \neq 0$, find the $\theta(t)$ in terms of $\dot{\theta}(0)$ for small $t$.

Solution: (a) Because the small sphere rolls without slipping on the surface of the fixed sphere, we have

$$R\theta = r(\phi - \theta) \quad \text{or} \quad (R+r)\theta = r\phi. \qquad (1)$$

(b) The conservation of mechanical energy gives

$$(R+r)(1-\cos\theta)mg = \frac{1}{2}m\left[(R+r)\dot{\theta}\right]^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right)\dot{\phi}^2$$

From (1), we have $(R+r)\dot{\theta} = r\dot{\phi}$. Then

$$(R+r)(1-\cos\theta)mg = \frac{1}{2}m\left[(R+r)\dot{\theta}\right]^2 + \frac{2}{10}m\left[(R+r)\dot{\theta}\right]^2 = \frac{7}{10}mv_c^2$$

Thus $v_c = \sqrt{\dfrac{10g}{7}(R+r)(1-\cos\theta)}$.

(c) The sphere will fly off when $\dfrac{mv_c^2}{R+r} \geq mg\cos\theta$. That is

$$\frac{10}{7}(1-\cos\theta) \geq \cos\theta \quad \text{or} \quad \cos\theta \leq \frac{10}{17}, \quad \theta_{\max} = \arccos\frac{10}{17}.$$

(d) The equation of rotational motion of the sphere around its contact point can be written as

$$\left(\frac{2}{5}mr^2 + mr^2\right)\ddot{\phi} = mgr\sin\theta$$

Using the relation $(R+r)\ddot{\theta} = r\ddot{\phi}$, we have $\dfrac{7}{5}(R+r)\ddot{\theta} = g\sin\theta$. For small $t$, using the approximation $\sin\theta \approx \theta$, the equation can be rewritten as

$$\ddot{\theta} - \kappa^2\theta = 0, \quad \kappa = \sqrt{\frac{5g}{7(R+r)}}.$$

The general solution is then $\theta(t) = Ae^{\kappa t} + Be^{-\kappa t}$. Substituting the initial condition $\theta(t=0) = 0$ into the general solution, we have $A + B = 0$ or $B = -A$. Thus the solution can be written as

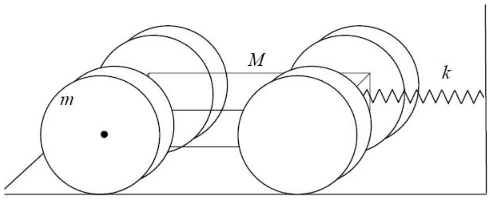
$$\theta(t) = A\left(e^{\kappa t} - e^{-\kappa t}\right). \qquad (2)$$

Substituting the initial condition $\dot{\theta}(t=0) = \dot{\theta}(0)$ into (2), we have

$$2A\kappa = \dot{\theta}(0) \quad \text{or} \quad A = \frac{\dot{\theta}(0)}{2\kappa}.$$

Therefore, $$\theta(t) = \frac{\dot{\theta}(0)}{2\kappa}\left(e^{\kappa t} - e^{-\kappa t}\right) = \frac{\dot{\theta}(0)}{\kappa}\sinh(\kappa t).$$

2. As shown in the figure, a toy car made of a rectangular block of mass $M$ and four disk wheels of mass $m$ and radius $R$ is placed on a horizontal table. One end of the car is attached to the vertical wall by a spring with spring constant $k$. Assume that the wheels of the car can roll without slipping on the table. Find the (circular) frequency of the small oscillations of the car.




Solution: The equation of translational motion of the toy car is

$$(M + 4m)\ddot{X} = -kX + 4f \tag{1}$$

where $f$ is the friction exerted on each wheel by the table. The equation of rotational motion of each wheel is

$$-fR = \frac{1}{2}mR^2\alpha . \tag{2}$$

Since the wheel does not slip on the table, we have the relation $R\alpha = \ddot{X}$. Substituting the relation into equation (2), we have

$$f = -\frac{1}{2}mR\alpha = -\frac{1}{2}m\ddot{X} .$$

Substituting $f$ into equation (1), we have $(M + 4m)\ddot{X} = -kX - 2m\ddot{X}$ or $\ddot{X} + \frac{k}{M + 6m}X = 0$ .

Thus,

$$\omega^2 = \frac{k}{M + 6m}, \quad \omega = \sqrt{\frac{k}{M + 6m}} .$$

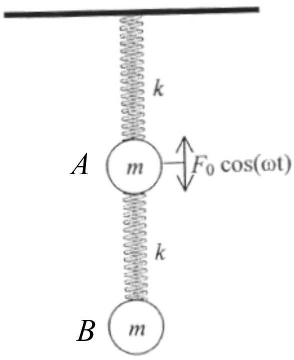
3. Two masses $A$ and $B$ both of mass $m$ are connected to two massless springs both of spring constant $k$ and are hanged vertically as shown in the figure. A vertical harmonic force $F(t) = F_0 \cos\omega t$ is applied to the mass $A$.

(a) Write the equations of motion of the two masses.

(b) Find the steady-state motion for each mass. Write the results in terms of $F_0$, $\omega$ and $\omega_0 = \sqrt{k/m}$ .

(c) If the circular frequency of the applied force is $\omega = \omega_0 = \sqrt{k/m}$ , what is the steady-state motion of each mass?

(d) Explain why the mass $A$ moves in this way in the case of (c).




Solution: (a) Taking the equilibrium positions of two masses as the origins of the coordinates, the equations of motion can be written as

$$\begin{aligned} m\ddot{x}_1 &= -kx_1 + k(x_2 - x_1) + F_0 \cos\omega t, \\ m\ddot{x}_2 &= -k(x_2 - x_1) \end{aligned} \tag{1}$$

where $x_1$ and $x_2$ denote the positions of mass $A$ and mass $B$ respectively.

(b) Equation (1) can be rewritten as

$$\begin{aligned} \ddot{x}_1 + 2\omega_0^2 x_1 - \omega_0^2 x_2 &= \frac{F_0}{m}\cos\omega t, \\ \ddot{x}_2 - \omega_0^2 x_1 + \omega_0^2 x_2 &= 0 \end{aligned} \tag{2}$$

Writing the steady-state oscillations of the mass $A$ and mass $B$ as $x_1(t) = A\cos(\omega t + \phi_1)$ and $x_2(t) = B\cos(\omega t + \phi_2)$, and substituting them into (2), we have

$$\left(-\omega^2 A\cos\phi_1 + 2\omega_0^2 A\cos\phi_1 - \omega_0^2 B\cos\phi_2\right)\cos\omega t - \left(-\omega^2 A\sin\phi_1 + 2\omega_0^2 A\sin\phi_1 - \omega_0^2 B\sin\phi_2\right)\sin\omega t = \frac{F_0}{m}\cos\omega t,$$
$$\left(-\omega^2 B\cos\phi_2 - \omega_0^2 A\cos\phi_1 + \omega_0^2 B\cos\phi_2\right)\cos\omega t - \left(-\omega^2 B\sin\phi_2 - \omega_0^2 A\sin\phi_1 + \omega_0^2 B\sin\phi_2\right)\sin\omega t = 0.$$

Obviously, we have $\phi_1 = \phi_2 = 0$ and

$$\left(2\omega_0^2-\omega^2\right)A-\omega_0^2 B=\frac{F_0}{m},$$
$$-\omega_0^2 A+\left(\omega_0^2-\omega^2\right)B=0.$$

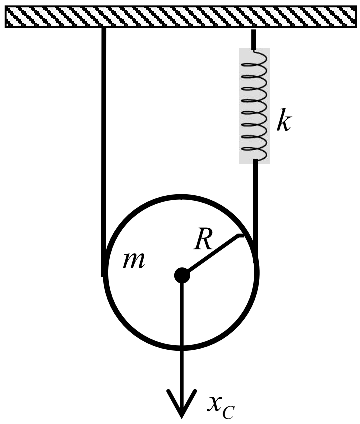
Thus, we obtain

$$A=\frac{F_0\left(\omega_0^2-\omega^2\right)}{m\left[\left(2\omega_0^2-\omega^2\right)\left(\omega_0^2-\omega^2\right)-\omega_0^4\right]},\qquad B=\frac{F_0\omega_0^2}{m\left[\left(2\omega_0^2-\omega^2\right)\left(\omega_0^2-\omega^2\right)-\omega_0^4\right]}.$$

(c) When $\omega=\omega_0=\sqrt{k/m}$, we have $A=0$ and $B=\dfrac{F_0}{m\omega_0^2}=\dfrac{F_0}{k}$. That is, the mass $A$ stays at rest and mass $B$ oscillates with amplitude of $\dfrac{F_0}{k}$.

(d) The mass $A$ moves in this way (staying at rest) because the external force acted on it is just canceled by the force exerted by the lower spring.

4. As shown in the accompanying figure, a cylinder with mass $m$ and radius $R$ is suspended from the ceiling with a soft massless rope round it. The left side of the rope is attached to the ceiling and the right side of the rope is attached to a spring of spring constant $k$ that is attached to the ceiling. There is sufficient friction between the rope and the cylinder to prevent the cylinder to slide on the rope. Find the oscillation frequency of the suspended cylinder.




Solution: Take the center of the cylinder when the spring is relaxed as the origin of the center of mass coordinates of the cylinder. Then, we have $x_C=x/2$, where $x_C$ is the coordinate of the center of mass of the cylinder and $x$ is the extended length of the spring. The equation of translational motion is

$$m\frac{d^2x_C}{dt^2}=mg-kx-T \qquad (1)$$

where $T$ is the tensional force of the left side of the rope. The rotational equation of motion is

$$\frac{1}{2}mR^2\frac{d^2\theta}{dt^2}=TR-kxR. \qquad (2)$$

The condition that the cylinder does not slide on the rope requires

$$\frac{dx_C}{dt}=\frac{d\theta}{dt}R. \qquad (3)$$

Substituting (3) into (2), we obtain $T=kx+\dfrac{1}{2}m\dfrac{d^2x_C}{dt^2}$. Substituting this result in to (2), we have

$$m\frac{d^2x_C}{dt^2}=mg-2kx-\frac{1}{2}m\frac{d^2x_C}{dt^2} \quad \text{or} \quad \frac{3}{2}m\frac{d^2x_C}{dt^2}=mg-4kx_C=-4k\left(x_C-\frac{mg}{4k}\right).$$

Let $\tilde{x}_C=x_C-\dfrac{mg}{4k}$, the above equation can be rewritten as

$$\frac{3}{2}m\frac{d^2\tilde{x}_C}{dt^2}=-4k\tilde{x}_C \quad \text{or} \quad \frac{d^2\tilde{x}_C}{dt^2}+\frac{8k}{3m}\tilde{x}_C=0.$$

Therefore, $\omega=\sqrt{\dfrac{8k}{3m}}$.